TECHNICAL NOTE — EXTENSION

Gradient Boosting and the Burt Representation

Does the representation-geometry effect survive non-linear pricing?

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Abstract. The companion technical note concluded that the discrimination gap between classical one-hot and Burt-coordinate representations is a property of the Tweedie GLM, and predicted that it would narrow under non-linear pricing. This extension tests that prediction directly. Replacing the Tweedie GLM with a gradient-boosted Tweedie learner — and holding the dataset, the Burt geometry, and the Gaussian-mixture segmentation fixed, with discrimination measured out-of-fold — the gap narrows by roughly one fifth but does not collapse: four fifths of it survives. The informational-completeness of the Burt coordinates persists into the non-linear setting, and the aggregate capital findings — the lower ground-up tail and the reinsurance-dependent coverage-layer reversal — are reproduced almost unchanged. The discrimination effect is therefore partly, not wholly, an artefact of the linear model; the capital-structure findings appear to be properties of the representation geometry itself.

1. The prediction under test

The companion note (May 2026) reported that enriched Gaussian-mixture segmentation on a classical one-hot geometry (experiment A5) achieves the strongest pricing discrimination, while the Burt-replacing geometry (B5) gives up roughly 4.5 Gini points but produces better top-decile calibration and a lower aggregate tail. It attributed the discrimination gap to the Tweedie GLM, "architecturally optimised for categorical contrasts," and predicted the gap would narrow under a non-linear learner.

That sentence is a falsifiable claim, and this note is the experiment that tests it. The design principle is minimal: change exactly one component of the pipeline — the pricing model — and leave everything else identical, so that any movement in the gap is attributable to the learner rather than to a change of features or segments.

2. Method

The Tweedie GLM is replaced by a gradient-boosted regressor with a Tweedie objective and variance power 1.5, matching the GLM's distributional family so that the comparison reflects the learner and not a change of loss function. The Burt construction, the PCA stabilisation, the Gaussian-mixture segmentation and its BIC selection, the empirical per-segment severity coefficients of variation, and the FFT aggregate machinery are all carried over unchanged from the original notebook.

Two methodological points are essential to the validity of the comparison. First, premiums are produced *out-of-fold*: each policy is priced by a model trained on data excluding it, through five-fold cross-fitting with a fixed partition shared across experiments. A gradient booster's in-sample discrimination is largely memorisation; an in-sample Gini would inflate every experiment and make the gap meaningless. Second, the boosting hyper-parameters are frozen across all experiments, so the contrast reflects representation geometry rather than uneven tuning effort.

One consequence of the non-linear setting deserves note. A Tweedie GLM fitted by maximum likelihood is mean-unbiased by construction; a Tweedie booster, descending on the log-mean scale, is not, and its back-transformed predictions carry a modest systematic level bias (the familiar Jensen gap). This is corrected by a single global calibration factor applied before any downstream metric. Because the Gini is rank-based, it is invariant to this global scaling, and the calibrated premiums reconcile to the portfolio total exactly, as in the linear case.

3. Results

3.1 Discrimination. Under the booster the A5 Gini falls from 0.4241 to 0.3958 and the B5 Gini from 0.3796 to 0.3615. The A5–B5 gap narrows from 0.0445 to 0.0346 — a reduction of approximately 23%. The result is stable across two independent fold partitions, agreeing to the fourth decimal. The prediction of the original note is therefore confirmed in direction but refuted in its stronger form: the gap narrows, but roughly four fifths of it survives. A meaningful part of the classical representation's advantage is not an artefact of the linear model; it reflects sharp, idiosyncratic categorical signal that the Burt embedding necessarily smooths into continuous coordinates.

Experiment	Gini GLM	Gini GBM	Δ Gini	Top-decile GBM
A5 classical, k=7	0.4241	0.3958	−0.0283	0.99
B5 Burt, k=8	0.3796	0.3615	−0.0181	0.91 / 0.63 *
C Burt-augmented (enriched)	0.3815	0.3660	−0.0155	0.92
C Burt-augmented (joint)	0.3781	0.3594	−0.0187	0.93

* The Burt top-decile ratio is partition-sensitive (see §4). All other figures are stable across partitions.

3.2 Informational completeness. The two Burt-augmented (C-series) designs — Burt coordinates alone, and Burt coordinates retained alongside the original categorical dummies — produce near-identical Gini under the booster, within 0.005, across both partitions. This is a stronger statement than its GLM-era counterpart: a tree is free to exploit the raw one-hot splits and the smooth coordinates simultaneously, and it finds essentially nothing in the dummies beyond what the Burt coordinates already encode. The completeness of the Burt representation is not an artefact of linear pricing.

3.3 Aggregate tail. Feeding the booster's segment-level parameters into the unchanged FFT aggregate machinery, the Burt portfolio retains its lower ground-up tail: a 99% TVaR approximately €75,000 below the classical portfolio, against €85,000 under the GLM, with a lower aggregate coefficient of variation (0.00991 vs 0.01038). Because portfolio mean and total claim frequency are held identical by construction, the entire difference resides in the higher moments — the Burt geometry distributes the same expected loss across its segments with lower dispersion and skew. The figures are near-invariant across partitions, consistent with the advantage arising from the segmentation geometry rather than the pricing model.

3.4 The coverage-layer reversal. The reinsurance-dependent reversal — the most novel finding of the original note — is reproduced under the booster, with all three signs stable across both partitions.

Coverage structure	GLM (B5 – A5)	GBM (B5 – A5)
Ground-up TVaR 99%	−€85k	−€75k
Deductible 150 — TVaR 99%	−€550k	−€517k
Per-claim limit 12k — TVaR 99%	+€478k	+€400k

Burt produces the lower 99% TVaR on a ground-up basis and under a 150 EUR deductible, while the classical representation produces the lower TVaR under a 12,000 EUR per-claim limit, which caps the larger individual claims driving Burt's heavier severity tail. The magnitudes attenuate modestly under the booster; the qualitative conclusion — that the preferred representation depends on the treaty structure in place — holds under non-linear pricing.

4. A caveat: partition-sensitive top-decile calibration

One metric does not carry over cleanly. The classical top-decile calibration ratio is stable across partitions (near 0.99–1.00); the Burt ratio is not, ranging from 0.63 to 0.91 across the two fold partitions tested. The Burt premium distribution is smoother and more compressed, so the boundary of its richest decile is less sharply defined and the policies falling just within it shift materially with the fold split. Under non-linear pricing the top-decile calibration of the Burt representation should therefore be reported as a distribution across partitions rather than as a single point estimate. This sensitivity does not affect the Gini gap, which is rank-based and stable, nor the aggregate findings, which derive from segment structure rather than the decile cut.

5. Bottom-up versus a directly-fitted collective model

A reviewer of the companion study observed that the aggregate analysis here is *bottom-up* — individual segment-level models aggregated upward into the portfolio distribution — whereas solvency analysis conventionally uses a *top-down* collective model fitted directly on portfolio data, with the individual model reserved for allocating cost downward. The two should coincide in theory and diverge in practice. It is worth testing the divergence directly, and being precise about its sources.

The divergence has two distinct origins, only one of which this benchmark can express. The first is **claims development**: a collective model fitted on reserved portfolio data reflects open claims — provisioned but not yet paid — that a model built on observed paid amounts does not. freMTPL2 is a settled, one-year benchmark with no such development to miss, so this source is absent here by construction; it is a property of production portfolios and is not modelled. The second is the **segmentation mixture**: the top-down severity is a single distribution fitted to the pooled claim amounts, while the bottom-up severity is a mixture of per-segment laws with heterogeneous coefficients of variation. Pooling discards that heterogeneity, and any divergence on this benchmark is attributable to it.

To isolate that effect, a collective model was fitted directly on the observed claim amounts — a single lognormal severity (CV 1.57) and the portfolio claim intensity — with its mean rescaled so that total expected loss and frequency match the bottom-up aggregates exactly. The only remaining difference between the two constructions is then the severity representation: pooled versus segmented.

Metric (99%)	Top-down	A5 bottom-up	B5 bottom-up
Mean	€59.91M	€59.91M	€59.91M
VaR	€61.29M	€61.37M	€61.31M
TVaR	€61.50M	€61.59M	€61.52M

Bottom-up TVaR 99% exceeds the directly-fitted collective by €94,700 (+0.15%) for A5 and €19,600 (+0.03%) for B5.

Two observations follow. First, on a settled benchmark the two approaches very nearly agree — to within 0.15% on the 99% TVaR — confirming that they coincide in the absence of claims development, and locating the larger divergence the reviewer noted in production portfolios rather than in the method itself. Second, the small residual is the segmentation-mixture effect, and it has a consistent direction: the bottom-up construction produces the heavier tail, because pooling washes out the high-dispersion segments that drive extreme losses. A directly-fitted collective model therefore mildly *understates* capital relative to the segmented construction on this portfolio.

The effect also orders by representation. The classical segmentation (A5), whose segment severities are more dispersed, departs from the pooled model almost five times as far as the smoother Burt segmentation (B5). The bottom-up/top-down gap is thus itself a function of representation geometry — the same structure that governs the discrimination and coverage-layer results elsewhere in this work — though on this benchmark its magnitude is modest.

6. Discussion

Read together, the findings separate two effects that the original note treated as one. The discrimination gap is partly a property of the linear model and narrows under a more flexible learner; the capital-structure findings — the lower aggregate tail and the coverage-layer reversal — are reproduced almost unchanged and appear to be properties of the representation geometry itself, mediated by how the Gaussian-mixture segmentation distributes severity heterogeneity. A natural way to state the conclusion is that representation geometry and pricing model act on different axes: the choice of learner moves the discrimination axis but leaves the aggregate-capital axis largely intact.

7. Conclusion

The prediction of the companion note is confirmed in direction and refined in substance. Under a non-linear Tweedie learner, the classical-versus-Burt discrimination gap narrows by about a fifth but four fifths survives; the informational completeness of the Burt coordinates persists; and the aggregate capital findings — the lower ground-up tail and the reinsurance-dependent reversal — are reproduced almost unchanged across two fold partitions. The capital-structure behaviour of the Burt representation is robust to the pricing model; the discrimination gap is partly a property of the linear model; and top-decile calibration is the single metric requiring a partition caveat. The most durable observation of the original work — that representation geometry propagates into aggregate and capital behaviour, not merely into pricing discrimination — survives the move to non-linear pricing.

As in the original study, I would welcome perspectives on whether similar interactions between representation geometry and collective-risk structure have been examined elsewhere in the actuarial literature.